

Denoising Monte Carlo Renders With Diffusion Models

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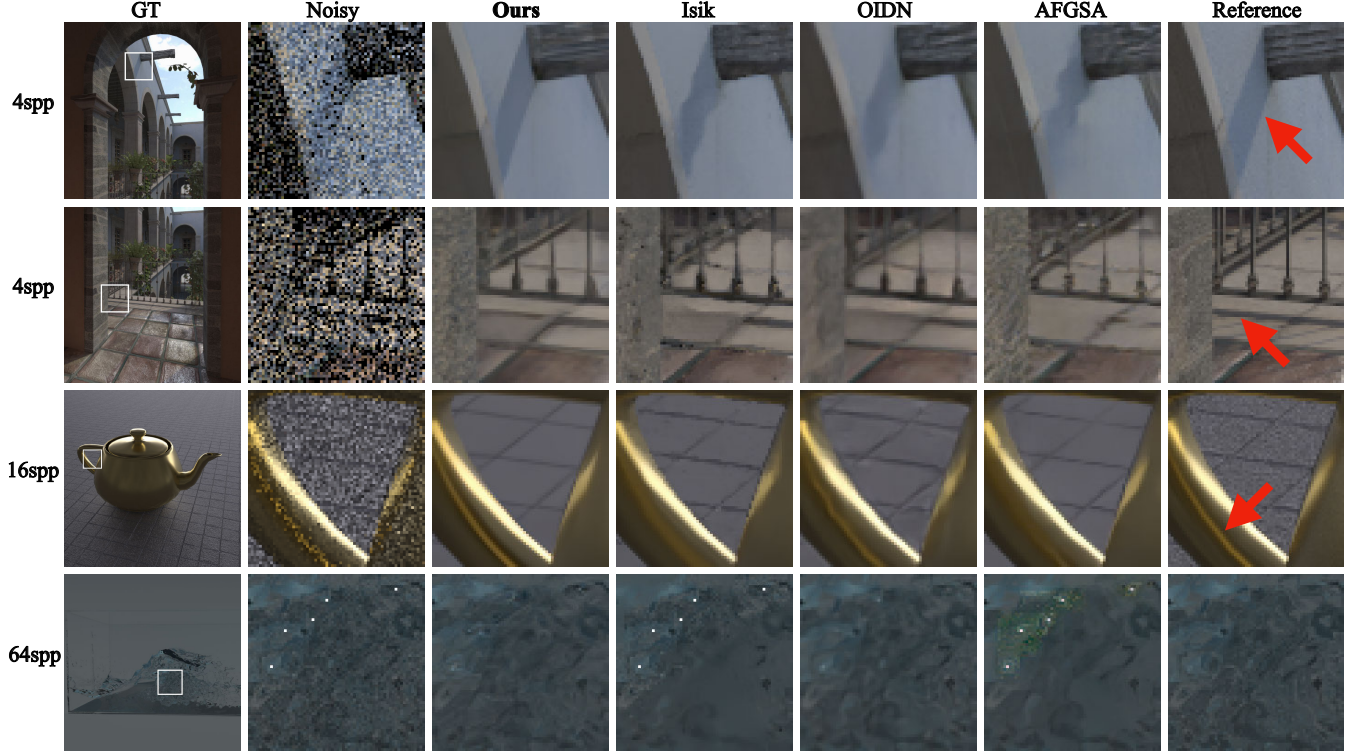


Figure 1: Our diffusion-based method can denoise MC renders in a plausible way.

ABSTRACT

Physically-based renderings contain Monte-Carlo noise, with variance that increases as the number of rays per pixel decreases. This noise, while zero-mean for good modern renderers, can have heavy tails (most notably, for scenes containing specular or refractive objects). Learned methods for restoring low fidelity renders are highly developed, because suppressing render noise means one can save compute and use fast renders with few rays per pixel. We demonstrate that a diffusion model can denoise low fidelity renders successfully. Furthermore, our method can be conditioned on a variety of natural render information, and this conditioning helps performance. Quantitative experiments show that our

method is competitive with SOTA across a range of sampling rates; qualitative evidence suggests that the image prior applied by a diffusion method strongly favors reconstructions that are like real images, with straight shadow boundaries, curved specularities, and no fireflies. In contrast, existing methods that do not rely on image foundation models struggle to generalize when pushed outside the training distribution.

CCS CONCEPTS

• Computing methodologies → Rendering; Reconstruction.

KEYWORDS

Monte Carlo Denoising, Diffusion Models

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Table 1: Quantitatively, our method is competitive with SOTA. We compare error metrics (L1, SMAPE, PSNR) across different spp settings (4, 16, 64) for various methods on our test set, consisting of 226 sequences of 8 frames each at 256 res. We remark that Isik’s method benefits from its ability to gather temporal information; all other methods are single-frame. All evaluation metrics only take into account single-frame performance. Our method outcompetes all others on L1, which is computed in HDR $[0 - 6]$ space, and is in-line with other methods in terms of PSNR. In the fourth row, we disable our Control Module and denoise with an off-the-shelf DeepFloyd Stage II model, presenting the best numbers after ablating on the number of skipped denoising time steps and other hyperparameters. Applying pure DeepFloyd to low-ray estimates is not successful. (Final row) Thus, conditioning our method with render buffer information via our Control Module makes a significant difference to performance.

Method	PSNR $\uparrow$			L1 $\downarrow$			SMAPE $\downarrow$		
	4spp	16spp	64spp	4spp	16spp	64spp	4spp	16spp	64spp
OIDN [Áfra 2024]	36.24	39.81	42.54	0.0643	0.0538	0.0491	0.0718	0.0466	0.0339
AFGSA [Yu et al. 2021]	38.69	42.07	45.09	0.0279	0.0184	0.0142	0.0538	0.0365	0.0267
Isik [Isik et al. 2021]	38.97	42.14	45.22	0.0525	0.0446	0.0426	0.0439	0.0309	0.0226
DeepFloyd-II [AI 2024]	26.67	27.34	30.84	0.1192	0.0959	0.0584	0.2314	0.2117	0.1384
Ours	38.68	41.62	44.24	0.0251	0.0171	0.0123	0.0602	0.0438	0.0353

1 INTRODUCTION

We present a method to denoise Monte Carlo renderings via a pixel-space diffusion model. To do so, we introduce additional controls to an existing open-source model, DeepFloyd [AI 2024]. See the supplement for our architecture. While control mechanisms exist for latent-space models like Stable Diffusion, they have not existed for pixel-space models. As we show, such pixel-level control is required for MC Denoising. **Until now, the concept of foundation models has not been applied to denoising Monte Carlo (MC) renders.**

2 METHOD

We render a dataset of approx. 45k frames using PBRT at 4, 16, 64 sampling rates as well as a GT target. We train a Control Module that forces the frozen base diffusion model to respect the conditioning (specular, diffuse, albedo, depth, normals etc.). We apply log tonemapping to ensure high dynamic range information is preserved.

3 EVALUATION

We show **qualitatively** in Fig. 1 that our reconstructions look like real images, because diffusion models have a very strong notion of what an image looks like (they have seen a huge dataset) and because the conditioning buffers offer strong guidance (e.g. normals, albedo, and depth). In the final column, red arrows point to areas of interest. **First row.** Notice the straight edge on the shadow (ours; real images tend to have straight shadow edges) compared with blurred or blotchy edges (others). **Second row.** In undersampled regions, our method fills in the shadow underneath a railing (real images do not have incomplete shadows); other methods render a blurred or incomplete shadow. **Third row.** Notice the clean sharp highlight on the teacup handle, and smooth highlight boundaries (ours; real images have clean sharp highlight boundaries) compared with absent highlights and blotchy boundaries (others). Notice also aliasing effects in the background, most prominent for OIDN, and absent from our reconstruction. **Fourth row.** Fireflies, also known

as spike noise, are single very bright pixels, which do not occur in real images. It is rare that fireflies appear at 64 spp in our training set, so AFGSA and Isik fail to remove them. OIDN succeeds in removing them, likely because we use their pretrained model trained on a large dataset.

We show **quantitatively** in Table 1 that our method is in-line with the SOTA works on our test set. In fact, after assessing qualitative behavior, our results suggest that existing standard error metrics sometimes do not capture the fact that our procedure synthesizes textures that look like a real image when others do not.

Our model requires 3.8 seconds at inference time on a single A40 GPU, performing 50 DDIM steps at FP16 to denoise a 256-res image. Large-scale 1-step models like distilled diffusion models and GANs may enable significantly faster performance [Sauer et al. 2024].

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